

Question Bank Compilation

Question 1

Let $A = [a_{ij}]_{2 \times 2}$ be a matrix such that $\text{Tr}(A) = 5$.

If $f(x) = \det(A - xI)$ and $f(A) = A^2 + aA + 3I$ then value of $\text{Tr}[(\text{adj } A)^2 + (a + 1)\text{adj } A + 3I]$ is equal to $[\text{Tr}(P)]$ is trace of matrix P]

A 0

B 1

C 7

D 5

Question 2

Using all the letters of the word EENADU, words are formed and arranged as in dictionary order, then the rank of "EENADU" is _____

A 100

B 181

C 182

D 183

Question 3

Let

$$(x^3 + px^2 + 2x - 5)^{19}(x^2 + qx - 41)^8(x^4 - x^3 + x - 7)^6 =$$

be an identity, where $p, q, a_{95} \dots a_0$ are integers. Compute the smallest positive value of p .



A 6

B 5

C 8

D 7

Question 4

Given two lines defined by their Cartesian equations:

$L_1 : \frac{x-2}{1} = \frac{y+1}{2} = \frac{z-3}{-1}$ and $L_2 : \frac{x+1}{2} = \frac{y-1}{\alpha} = \frac{z}{1}$. If the shortest distance separating these lines is $\frac{3}{\sqrt{14}}$, find the sum of all permissible values of α .

A $\frac{2}{27}$

B $-\frac{2}{27}$

C $\frac{5}{54}$

D $-\frac{5}{54}$

Question 5

Box A contains 2 black and 3 red balls while box B contains 3 black and 4 red balls. Out of these two boxes one is selected at random and the probability of choosing box A is double that of box B . If a red ball is drawn from the selected box, then the probability that it has come from box B is:

A $\frac{21}{41}$

B $\frac{10}{31}$

C $\frac{12}{31}$

D $\frac{13}{41}$

Question 6

If $f(x) = \int e^{\sin x} \left(\frac{5 \cos x + \cos^3 x}{(2 - \sin x) \sqrt{3 + \cos^2 x}} \right) dx$ and $f(0) = 1$ then value of $f(\pi/2)$ equals

A $\sqrt{3}e$

B $\frac{\sqrt{3}e}{2}$

C $\frac{3e}{2}$

D $3e$

Question 7

The area inside the parabola $5x^2 - y = 0$ but outside the parabola $2x^2 - y + 9 = 0$ is equal to: (sq. unit)

A $6\sqrt{3}$

B $12\sqrt{3}$

C $3\sqrt{3}$

D $24\sqrt{3}$

Question 8

Let n be the number of integral values of x , which satisfy the equation $\tan^{-1} x - \sin^{-1} x = \cos^{-1} x - \cot^{-1} x$.

Let $\vec{c} = |\vec{a} \times \vec{b}| \vec{a} + (\vec{a} \cdot \vec{b}) \vec{b}$ where $\vec{a}$ and $\vec{b}$ are unit vectors and angle between $\vec{a}$ and $\vec{b}$ is $\frac{\pi}{6}$

Let the value of definite integral $\int_{-\frac{\pi}{2}}^{\frac{\pi}{2}} \frac{\ln(\sec x)}{(2013)^{\sin x} + 1} dx$ equals $\frac{\pi}{2} \ln k$,

then the value of $(n + 2|\vec{c}|^2 + k)$ is equal to....

A 12

B $\frac{19}{4}$

C $\frac{17}{2}$

D None of these

Question 9

Given the parabola $x^2 = -4y$, a point $P(4, -4)$ lies on it. A focal chord PQ is drawn through P . Let M and N be the points where perpendiculars from P and Q intersect the directrix of the parabola. Find the area of the quadrilateral $PQMN$.

A $\frac{125}{8}$

B $\frac{25}{2}$

C $\frac{75}{4}$

D $\frac{100}{3}$

Question 10

If $\vec{a}$ and $\vec{b}$ are two vectors such that $|\vec{a}| = 1$, $|\vec{b}| = 4$, $\vec{a} \cdot \vec{b} = 2$. If $\vec{c} = (2\vec{a} \times \vec{b}) - 3\vec{b}$ then angle between $\vec{b}$ and $\vec{c}$ is :

A $\frac{\pi}{6}$

B $\frac{\pi}{3}$

C $\frac{2\pi}{3}$

D $\frac{5\pi}{6}$

Question 11

Let M be the value of the limit $\lim_{x \rightarrow \infty} \left(-\frac{3}{2} \left(\frac{1}{3} - \frac{x^2}{x^2+1} \right) \right)^{x^2}$.
Determine the value of the expression $\frac{\log_e M}{2 + \log_e M}$.

A $\frac{3}{2}$

B $-\frac{3}{2}$

C 3

D -3

Question 12

Let $S = \{1, 2, 3, 4, 5, 6, 7\}$. Then the number of possible functions $f : S \rightarrow S$ such that $f(m \cdot n) = f(m) \cdot f(n)$ for every $m, n \in S$ and $m \cdot n \in S$ is equal to _____.

A 360

B 460

C 390

D 490

Question 13

If the ratio of k^{th} terms of two A.P.'s are in the ratio $(2k - 3) : (3k + 1)$ and the ratio of their sum to first 65 terms of these two AP's is $a : b$ (where a and b are coprime numbers), then the value of $(b + 2a)$ is

A 246

B 226

C 248

D 228

Question 14

A line with direction ratios $(2, 7, -5)$ is drawn to intersect the lines $\frac{x-5}{3} = \frac{y-7}{-1} = \frac{z+2}{1}$ and $\frac{x+3}{-3} = \frac{y-3}{2} = \frac{z-6}{4}$ at P and Q respectively (length of PQ)² is _____

A 82

B 78

C 64

D 49

Question 15

Consider the three linear equations with respect to X, Y, Z as

$$(x)X + Y + 2Z = 0$$

$$(f(x))X + 3Y + x^2Z = 0$$

$$(5x)X + 6Y + Z = 0$$

having non-trivial solution for X, Y, Z . The curve $f(x)$:

A is always increasing

B is always decreasing

C has exactly one critical point

D has three critical points

Question 16

Solution $y = y(x)$ of the differential equation $(x^3 + 2xy)dy = (y^2 - yx^2)dx, y(1) = 1$ is given by -

A $2y^3 - x^3 = xy$

B $2x^2 + xy = 3y^3$

C $y^2 + x^2y = 2x$

D $x^2 - yx^2 = y^3$

Question 17

The side BC of a triangle ABC lies on x -axis. If its circumcentre is $(5, 7)$ and orthocentre is $(2, 3)$ and vertices $A = (a_1, b_1), B = (a_2, b_2), C = (a_3, b_3)$ (where $a_2 < a_3$) . Then

A $a_2 + a_3 = 12$

B $b_1 + b_2 + b_3 = 1$

C $B = (5 - \sqrt{30}, 0)$

D $C = (5 + \sqrt{60}, 0)$

Question 18

$$2(\cos x + \cos 2x) + (1 + 2 \cos x) \sin 2x = 2 \sin x$$

Total no. of solutions in $[-\pi, \pi]$ is _____

A 3

B 6

C 4

D 5

Question 19

If $A(z_1)$ and $B(z_2)$ be two complex numbers satisfying the equation $z + (\sqrt{3} + i)t - i = 0$ and $\sqrt{3}z + (1 + i\sqrt{3})\lambda - \sqrt{3}i = 0$ respectively $\forall t, \lambda \in \mathbb{R}$. Let for some z_1 and z_2 , $\arg(z_1) = \frac{\pi}{4}$ and $\arg(z_2) = -\frac{3\pi}{4}$, then the area of the triangle ABC (where C is point $0 + i$) is

A $\frac{\sqrt{3}+1}{2}$

B $\frac{\sqrt{3}+1}{4}$

C $\frac{\sqrt{3}+2}{4}$

D $\frac{\sqrt{3}+2}{2}$

Question 20

Let ellipse $\frac{x^2}{25} + \frac{y^2}{p^2} = 1$ and hyperbola $\frac{x^2}{9} - \frac{y^2}{q^2} = 1$ are confocal and meet at points A,B,C and D. If foci of ellipse subtend 90° at point A and e_1 and e_2 are eccentricities of ellipse and hyperbola then value of $25e_1^2 + 9e_2^2$ is

A 34

B 28

C 30

D 36

Question 21

What is the remainder when $((7)^{77})^{77}$ is divided by 10?

Question 22

Let $S = \{1, 2, 3, 4\}$. A binary relation P is defined on S such that for any $a, b \in S$, aPb if and only if $b = a + 1$. Let k be the total number of ordered pairs in P . Let p be the minimum number of additional ordered pairs required to make P a reflexive relation. Let q be the minimum number of additional ordered pairs required to make P a symmetric relation. Determine the value of $k + p + q$.

Question 23

The highest positive integral value of λ so that the equation $x^2 - 5|x| + 6 - |\lambda - 2| = 0$ has four distinct roots is _____.

Question 24

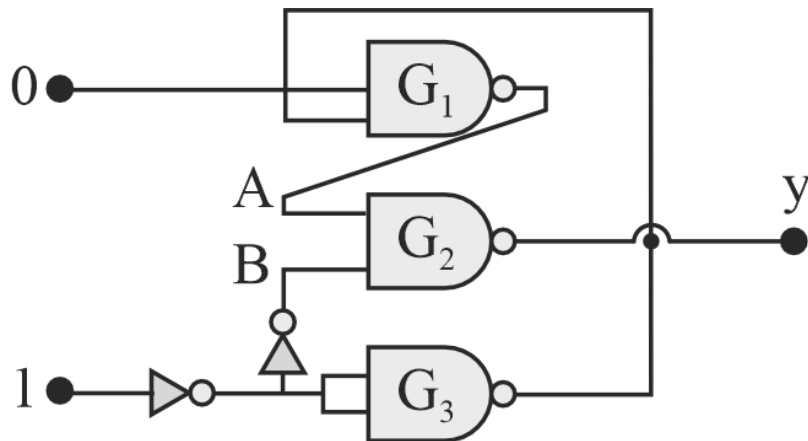
If $f: R^+ \rightarrow R$ be a differentiable function satisfying $f(x) = x \ln x + x - \int_1^x \frac{f(t)}{t \cdot \ln t} dt \forall x \in R^+$, then the value of $f'(e) + f''(\frac{1}{4}) =$

Question 25

The number of integral values of λ for which the variable line $3x + 4y - \lambda = 0$ lies between the circles $x^2 + y^2 - 2x - 2y + 1 = 0$ and $x^2 + y^2 - 18x - 2y + 78 = 0$ without having any common points with any circle is equal to $2k$, then value of k is--

Question 26

In circuit in following fig. the value of Y is



A 0

B 1

C Fluctuates between 0 and 1

D Indeterminate as the circuit can't be realised

Question 27

A metal plate of area 10^3 cm^2 rests on a layer of oil 6 mm thick. A tangential force of 10^{-2} N is applied on it to move it with a constant velocity of 6 cms^{-1} . The coefficient of viscosity of the liquid is

A 0.1 poise

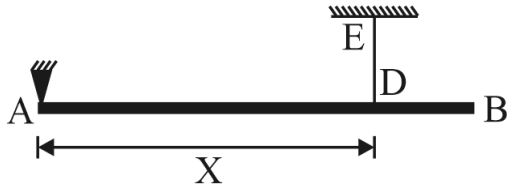
B 0.5 poise

C 0.7 poise

D 0.9 poise

Question 28

A uniform slender bar AB of length L is supported in horizontal position as in figure. At what distance x from the hinge A, should the vertical string DE be attached to the bar in order that, when it is cut, there will be no immediate change in the reaction at A



A $\frac{2L}{5}$

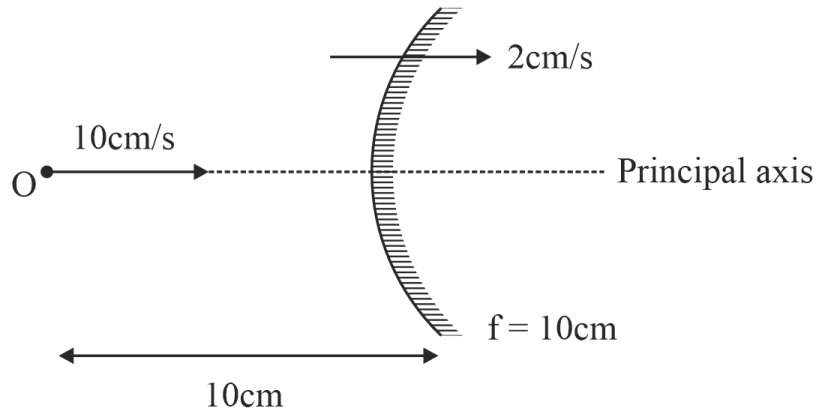
B $\frac{3L}{5}$

C $\frac{2L}{3}$

D $\frac{3L}{4}$

Question 29

What is the velocity (in cm/s) of image in situation shown below (O = object, f = focal length). Object moves with velocity 10cm/s and mirror moves with velocity 2 cm/s as shown



A 0

B 2

C 6

D 4

Question 30

Two point dipoles $p\hat{k}$ and $\frac{p}{2}\hat{k}$ are located at $(0, 0, 0)$ & $(1 \text{ m}, 0, 2 \text{ m})$ respectively. The resultant electric field due to the two dipoles at the point $(1 \text{ m}, 0, 0)$ is:

A $\frac{9p}{32\pi\epsilon_0} \hat{k}$

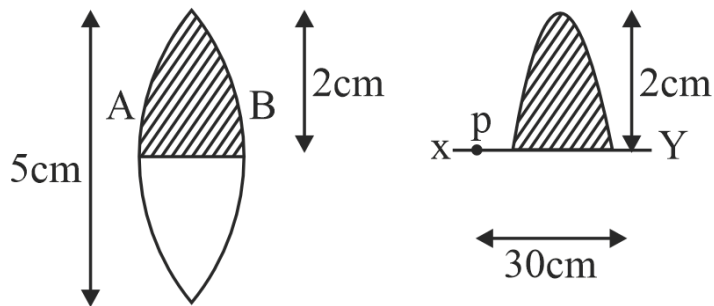
B $\frac{-7p}{32\pi\epsilon_0} \hat{k}$

C $\frac{7p}{32\pi\epsilon_0} \hat{k}$

D none of these

Question 31

A converging lens of focal length 20 cm and diameter 5 cm is cut along the line AB. The part of the lens shown shaded in the diagram is now used to form an image of a point P placed 30 cm away from it on the line XY. Which is perpendicular to the plane of the lens. The image of P will be formed.



A 0.5 cm above XY

B 1 cm below XY

C on XY

D 1.5 cm below XY

Question 32

Evaluate the following statements concerning thermodynamic properties:

- (I) An extensive property's value is directly proportional to the size or amount of the system.
- (II) Pressure (P), temperature (T), and specific volume (v) are examples of intensive properties.
- (III) Internal energy (U), entropy (S), and number of moles (n) are intensive properties.

Choose the option with the correct statement(s).

A (I) and (II) only

B (II) and (III) only

C (I) only

D (I), (II) and (III)

Question 33

A thin, non-conducting disk of radius R carries a uniform surface charge density σ . The disk rotates about its central axis with a constant angular velocity Ω . An Amperian loop P_1 is conceptually defined to measure the total effective current produced by this rotating charged disk. A second Amperian loop P_2 is a large circular path, concentric with the disk, and completely surrounds it. Determine the value of $I_{P_1} - I_{P_2}$, where I_{P_1} is the total effective current and I_{P_2} is the net current passing through the area bounded by loop P_2 .

A $\sigma\Omega R^2$

B $\frac{1}{2} \sigma\Omega R^2$

C $\frac{1}{4} \sigma\Omega R^2$

D 0

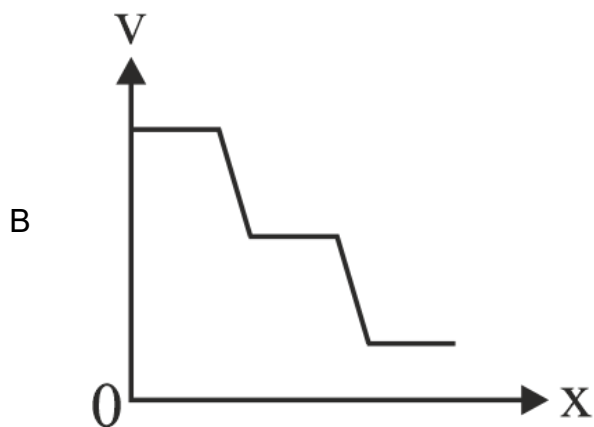
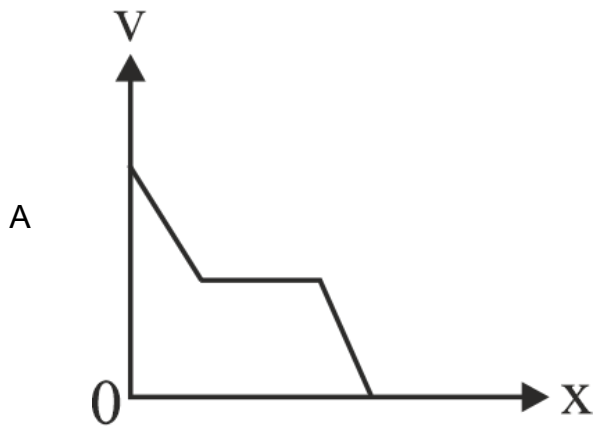
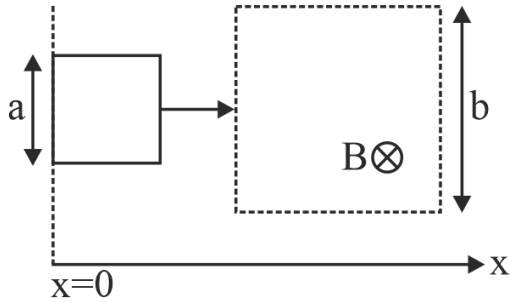
Question 34

An experimenter performs Young's double-slit experiment using two different monochromatic light sources: one emitting green light and another emitting violet light. Which of the following statements accurately describes the comparison of the interference patterns produced by these two light sources, assuming all other experimental parameters are kept constant?

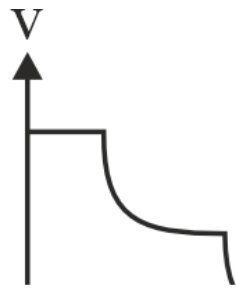
- A The fringe width produced by green light will be smaller than that produced by violet light.
- B The fringe width produced by green light will be larger than that produced by violet light.
- C The fringe width will be the same for both green and violet light, but the intensity will differ.
- D Violet light will produce a clearer interference pattern due to its higher frequency.

Question 35

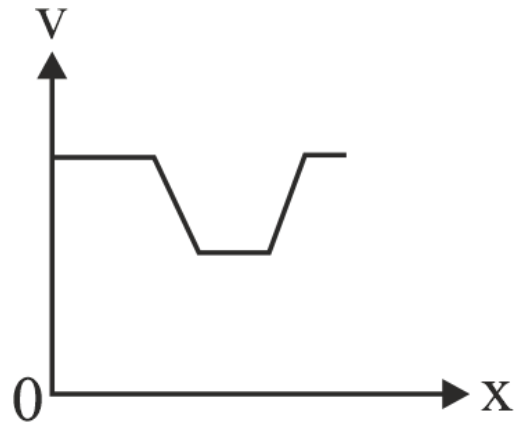
A square-shaped conducting wire loop of dimension a moving parallel to the x -axis approaches a square region of size b ($a < b$) where a uniform magnetic field B exists pointing into the plane of the paper (see figure). As the loop passes through this region, the plot correctly depicting its speed (v) as a function of x is



C



D



Question 36

A particle is projected from ground towards a vertical wall 80 m away at an angle of 37° with horizontal with initial velocity of 50 m/s . After its collision with wall & then once with ground find at what distance in meter from wall will it strike the ground. The component of velocity normal to the surface becomes half after collision with each surface.

A 60 m

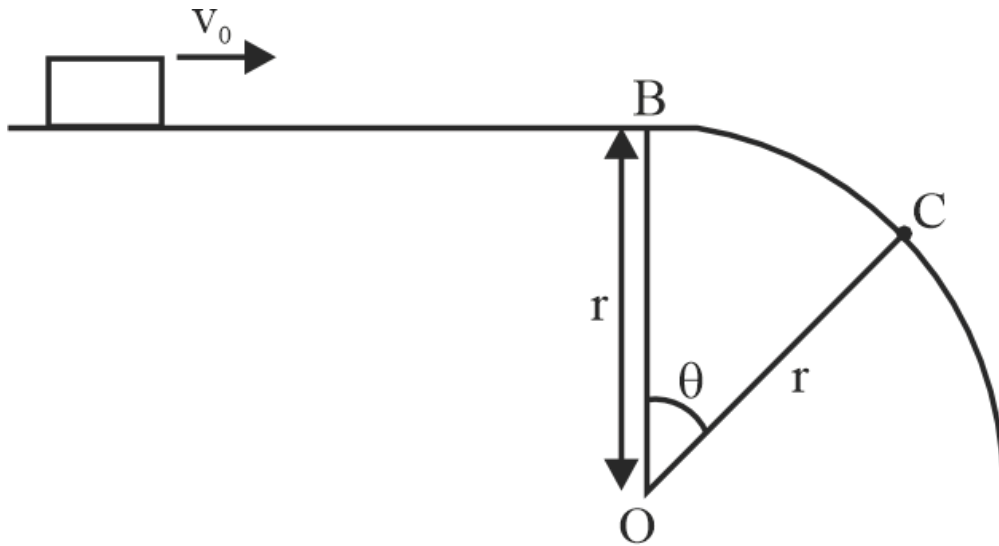
B 80 m

C 140 m

D 160 m

Question 37

A small block slides with velocity $0.5 \sqrt{gr}$ on the horizontal frictionless surface which after a point B is in a circular path in the vertical plane as shown in the figure. The block leaves the surface at the point C where it makes angle θ from vertical. The value of the angle θ is:



A $\cos^{-1} \left(\frac{4}{9} \right)$.

B $\cos^{-1} \left(\frac{3}{5} \right)$.

C $\cos^{-1} \left(\frac{1}{2} \right)$.

D None of the above.

Question 38

A mass M is suspended from a spring of negligible mass. The spring is pulled a little and then released so that the mass executes SHM of time period T . If the mass is increased by m , the time period becomes $5T/3$, then the ratio of $\frac{m}{M}$ is :

A $\frac{3}{5}$

B $\frac{25}{9}$

C $\frac{16}{9}$

D $\frac{5}{3}$

Question 39

In a series LCR the voltage across resistance, capacitance and inductance is 10 V each. If the capacitance is short circuited, the voltage across the inductance will be....

A $\frac{10}{\sqrt{2}}$ V

B 10 V

C $10\sqrt{2}$ V

D 20 V

Question 40

Consider the following physical quantities and their corresponding dimensional formulas. Identify the correct match from the given choices.

List-I (Physical Quantity)	List-II (Dimensional Formula)
(A) Momentum	(I) $[M^0 L^0 T^0]$
(B) Strain	(II) $[MLT^{-1}]$
(C) Gravitational Constant (G)	(III) $[M^{-1} L^3 T^{-2}]$
(D) Planck's Constant (h)	(IV) $[ML^2 T^{-1}]$

Select the option that correctly pairs the items from List-I with List-II:



A (A)-(II), (B)-(I), (C)-(III), (D)-(IV)

B (A)-(I), (B)-(II), (C)-(IV), (D)-(III)

C (A)-(II), (B)-(III), (C)-(I), (D)-(IV)

D (A)-(IV), (B)-(I), (C)-(III), (D)-(II)

Question 41

A long glass capillary tube of uniform bore of diameter 1 mm with both ends open is filled completely with water and then held vertically in air. If the surface tension of water is 0.075 N/m , the length of the water column remaining in the glass tube is ($g = 10 \text{ m/s}^2$)

A zero

B 1.5 cm

C 3.0 cm

D 6.0 cm

Question 42

When a metal plate is exposed to light of wavelength 400 nm, a negative potential of 1.1 V is found to be needed to stop the photo current. The threshold wavelength of the photoelectrons is.....

A 450 nm

B 500 nm

C 580 nm

D 620 nm

Question 43

A physical quantity K is determined from measurements of four other quantities A, B, C , and D using the relation $K = \frac{A^3 B}{C^2 \sqrt{D}}$. The percentage errors in the measurements of A, B, C , and D are 2%, 1%, 4%, and 3% respectively. What is the maximum percentage error in the calculated value of K ?

A 14.5%

B 18.5%

C 16.5%

D 15.5%

Question 44

During an adiabatic process, the density of a gas is found to be proportional to cube of temperature. The degree of freedom of gas molecule is....

A 6

B 5

C 4

D 3

Question 45

A particle moves along X -axis from $x = 0$ to $x = 5$ m under the influence of force given by $F = (7 - 2x + 3x^2)$ N . The work done in the process is.....

A 70 J

B 270 J

C 35 J

D 135 J

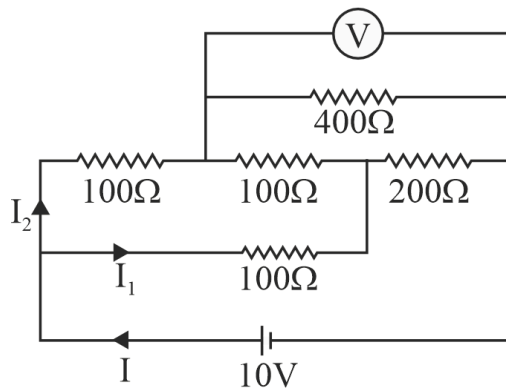
Question 46

A beam of unknown charged particles passes undeflected through a velocity selector where the electric field is $5 \times 10^4 \text{ N/C}$ and the magnetic field is 0.25 T . Both fields are uniform and mutually perpendicular, and also perpendicular to the particle's velocity. After exiting the selector, the particles enter a region containing only the same magnetic field (0.25 T) and follow a circular trajectory with a radius of 4 cm . Calculate the magnitude of the charge-to-mass ratio (q/m) for these particles in C/kg . Express your answer in the form $y \times 10^7 \text{ C/kg}$, and provide the value of y .

Question 47

An electrical circuit is shown in fig. The potential difference across the resistor of 400 ohm , as it is measured by the voltmeter V of resistance 400 ohm is $5k/3$.

Where $k =$ _____



Question 48

A parallel plate capacitor of capacitance $20\mu\text{ F}$ is being charged by a voltage source whose potential is changing at the rate of 3 V s^{-1} . The conduction current through the connecting wires, and the displacement current through the plates of the capacitor would be, respectively $N\mu\text{A}$ and $M\mu\text{A}$ find value of $M + 2N$.

Question 49

A long, slender pipe of uniform mass per unit length λ and total length L is filled with a light liquid and sealed. It is then rotated about one of its ends in a horizontal plane with a constant angular velocity ω . The tension in the pipe at a point located at a distance $L/4$ from the pivot is $\frac{p}{q} \lambda \omega^2 L^2$. The value of $p + q$ is _____. (p & q are coprime)

Question 50

An infinitely long cylindrical wire of radius ' R ' is carrying a current with current density $\vec{j} = \alpha r^3$ (where α is constant and ' r ' is the radial distance from the axis of the wire). If the magnetic field at $r = \frac{R}{2}$ is ' B_1 ' and at $r = 2R$ is ' B_2 ' then the ratio $\frac{B_2}{B_1}$ is

Question 51

Match List -I with List-II

List-I	List-II
(a) Lassaigne's test	(i) Carbon
(b) Cu (II) oxide	(ii) Sulphur
(c) Silver nitrate	(iii) N, S, P and halogen
(d) The sodium fusion extract gives black precipitate with acetic acid and lead acetate	(iv) Halogen specifically

The correct match is:

A $a \rightarrow iii; b \rightarrow i; c \rightarrow iv, d \rightarrow ii$

B $a \rightarrow i; b \rightarrow ii; c \rightarrow iv, d \rightarrow iii$

C $a \rightarrow i; b \rightarrow iv; c \rightarrow iii, d \rightarrow ii$

D $a \rightarrow iii; b \rightarrow i; c \rightarrow ii, d \rightarrow iv$

Question 52

A gaseous compound is composed of 85.7% by mass of carbon and 14.3% by mass of hydrogen. Its density is 1.95 g L^{-1} at 350 K and 1.0 atm pressure. Calculate the molecular formula of the compound.

A C_5H_{10}

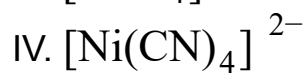
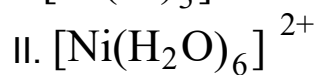
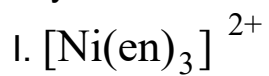
B C_6H_{12}

C C_4H_8

D C_8H_{18}

Question 53

Arrange the following nickel(II) complexes in ascending order of their Crystal Field Stabilization Energy (CFSE) values:



A $\text{III} < \text{II} < \text{I} < \text{IV}$

B $\text{II} < \text{III} < \text{I} < \text{IV}$

C $\text{III} < \text{I} < \text{II} < \text{IV}$

D $\text{I} < \text{II} < \text{III} < \text{IV}$

Question 54

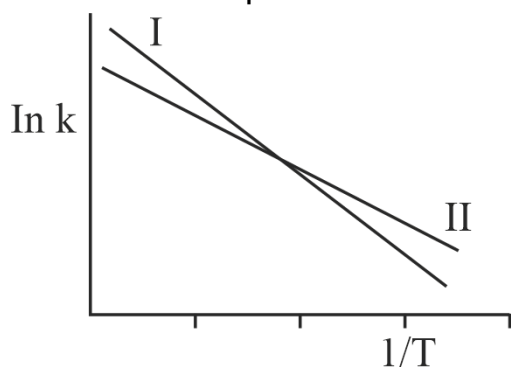
Assertion : Salts like KCl , KNO_3 i.e., inert electrolytes are used in salt bridge.

Reason : An inert electrolyte can easily be filled in the Utube.

- A If both assertion and reason are true and the reason is the correct explanation of the assertion.
 - B If both assertion and reason are true but reason is not the correct explanation of the assertion.
 - C If assertion is true but reason is false.
 - D If the assertion and reason both are false.
-

Question 55

The Arrhenius plots of two reactions, I and II are shown graphically:



The graph suggests that:

A $E_I > E_{II}$ and $A_I > A_{II}$

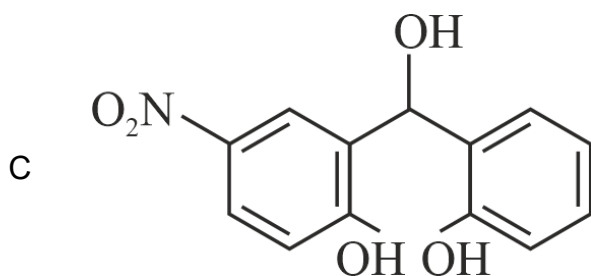
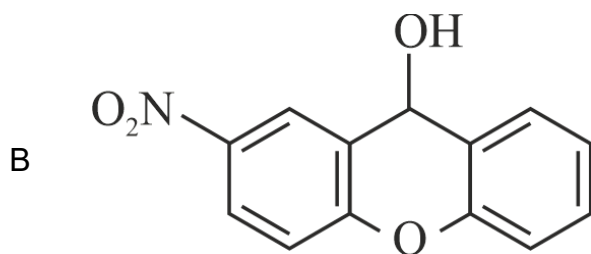
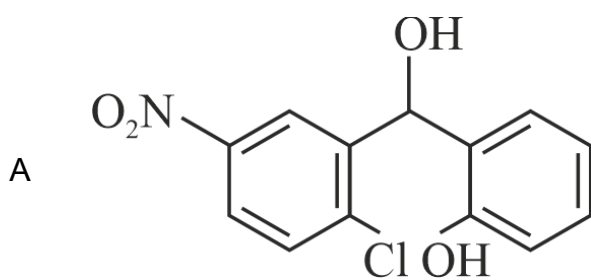
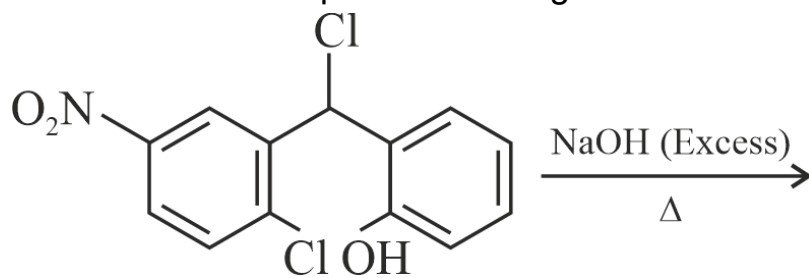
B $E_{II} > E_I$ and $A_{II} > A_I$

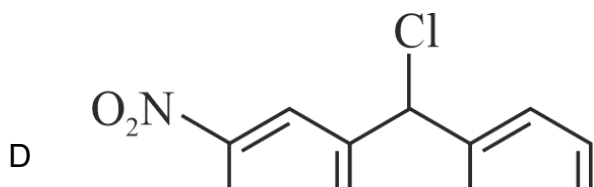
C $E_I > E_{II}$ and $A_{II} > A_I$

D $E_{II} > E_I$ and $A_I > A_{II}$

Question 56

What would be the product of the given reaction?





Question 57

The Mn^{3+} ion is unstable in solution and undergoes disproportionation to give Mn^{2+} , MnO_2 and H^+ ion. Write a balanced ionic equation for the reaction and stoichiometric coefficients for Mn^{3+} , MnO_2 and H^+ May be

A 2, 1, 4

B 3, 2, 6

C 3, 2, 4

D 2, 1, 3

Question 58

A hypothetical sparingly soluble compound, $\text{Ga}_3(\text{PO}_4)_5$, dissociates in water to form Ga^{5+} and PO_4^{3-} ions. If 's' denotes the molar solubility of this compound, determine the correct relationship between 's' and its solubility product constant, K_{sp} .

A $\left(\frac{K_{\text{sp}}}{84375}\right)^{\frac{1}{8}}$

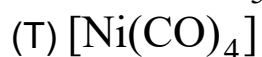
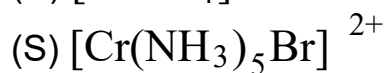
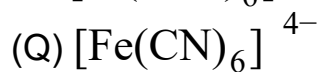
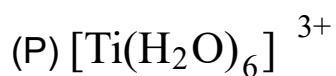
B $\left(\frac{K_{\text{sp}}}{15625}\right)^{\frac{1}{8}}$

C $\left(\frac{K_{\text{sp}}}{27}\right)^{\frac{1}{8}}$

D $\left(\frac{K_{\text{sp}}}{46875}\right)^{\frac{1}{8}}$

Question 59

Which of the given coordination compounds are classified as homoleptic and contain a central metal ion with an odd number of d electrons?



Select the appropriate choice:

A (P) and (R) only

B (Q) and (T) only

C (P), (Q) and (S) only

D (R) and (S) only

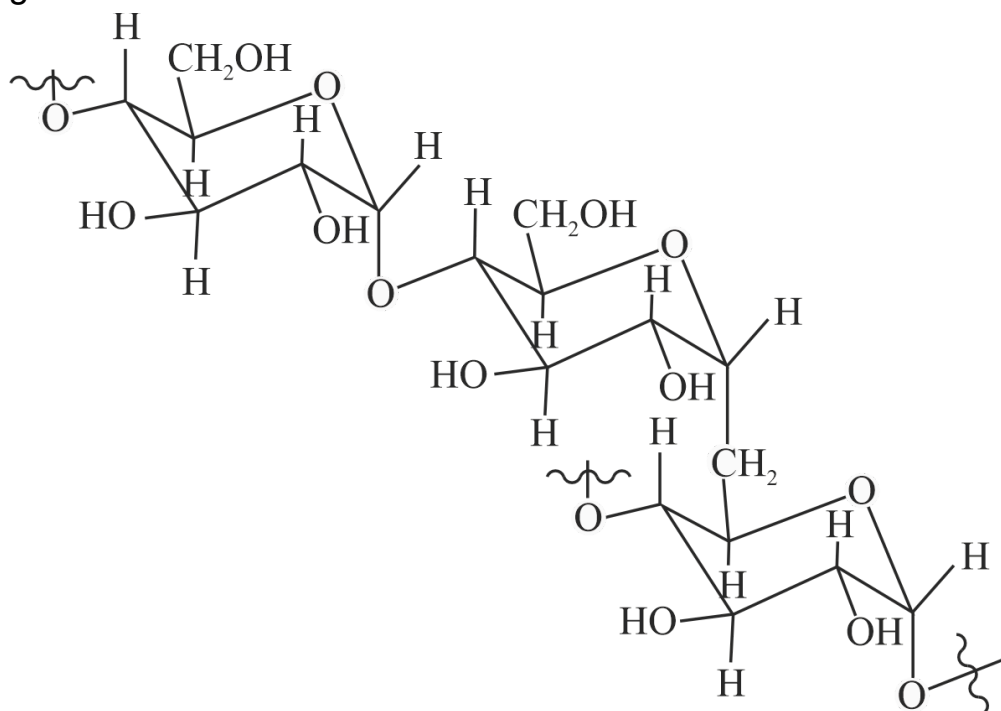
Question 60

Which of the following thermodynamic identities is INCORRECT?

- A The heat absorbed at constant pressure is equal to the change in enthalpy, i.e., $dq_P = dH$.
- B The heat absorbed at constant volume is equal to the change in internal energy, i.e., $dq_V = dU$.
- C For a system at constant temperature, the change in Gibbs free energy with pressure is given by $\left(\frac{\partial G}{\partial P}\right)_T = -V$.
- D The molar heat capacity at constant pressure is defined as $C_P = \left(\frac{\partial H}{\partial T}\right)_P$.

Question 61

Starch consist of amylose and amylopectin structure of amylopectin is given as below



In structure of amylopectin

A β -1,4-glycosidic bonds

B α - 1,6 -glycosidic bonds

C α - 1,4 - glycosidic bonds

D α - 1,4 -glycosidic bonds and α - 1,6 -glycosidic bonds

Question 62

For a hydrogen atom, which of the following sets of orbitals represents a degenerate group (i.e., all orbitals within the set have the same energy)?

A $2s, 2p_x, 3s$

B $3p_y, 3d_{xy}, 3s$

C $1s, 2s, 3s$

D $4s, 4p_z, 5d_{z^2}$

Question 63

The 1st, 2nd and the 3rd ionization enthalpies I_1 , I_2 and I_3 , of four atoms with atomic numbers n , $n + 1$, $n + 2$ and $n + 3$, where $n < 10$, are tabulated below. What is the value of n ?

Atomic Number	Ionization Enthalpy (kJ/mol)		
	I_1	I_2	I_3
n	1579	3374	6043
$n + 1$	2075	3952	6125
$n + 2$	493	4562	6907
$n + 3$	739	1451	7731

A 6

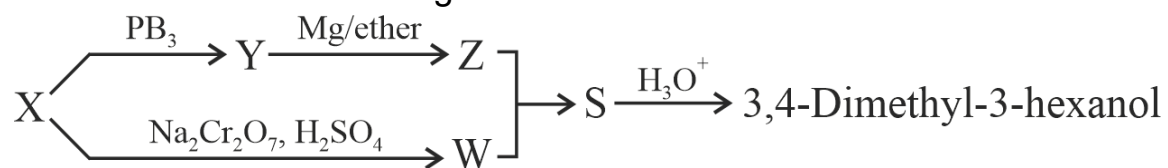
B 7

C 8

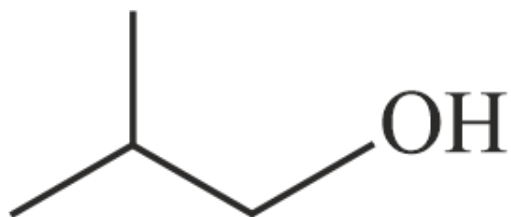
D 9

Question 64

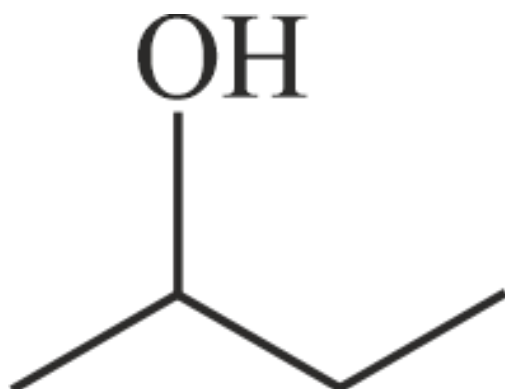
The structure of the starting material X is :



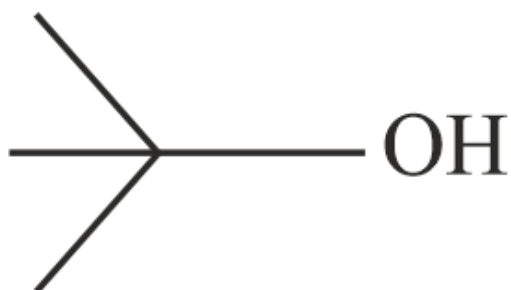
A



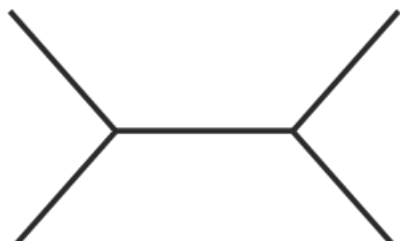
B



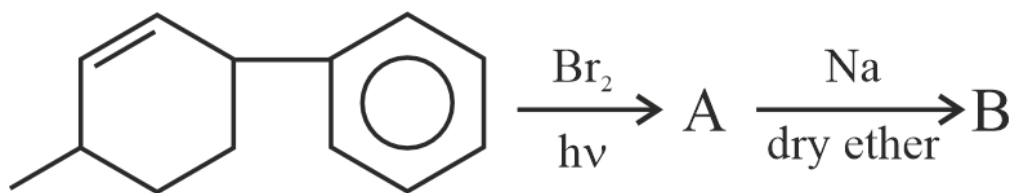
C



D

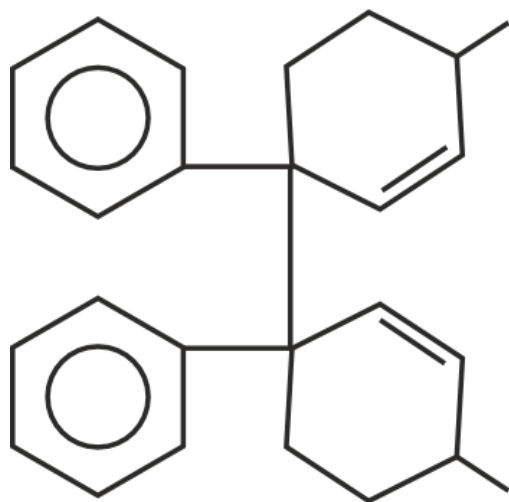


Question 65

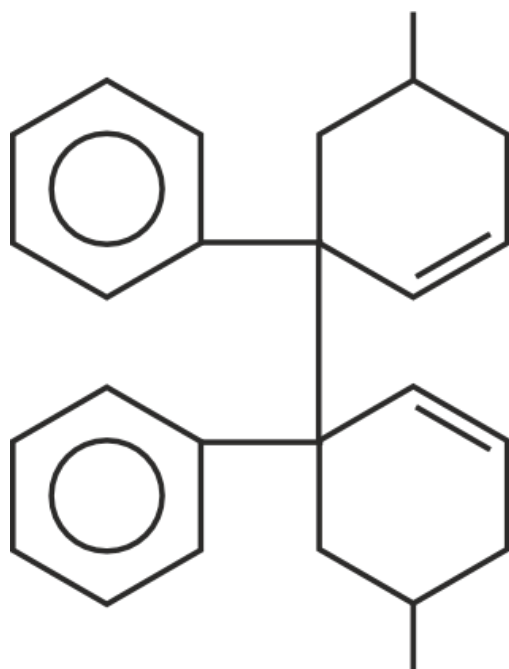


Major product *B* is_____

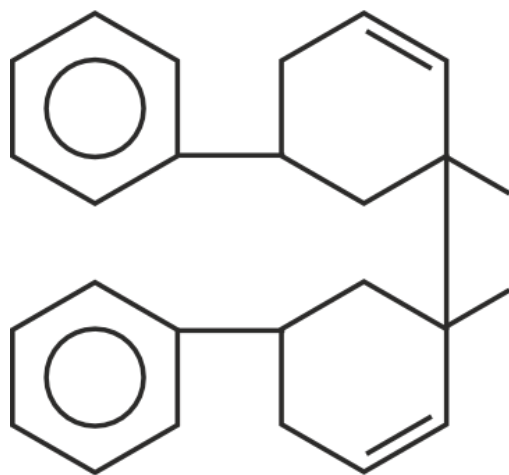
A



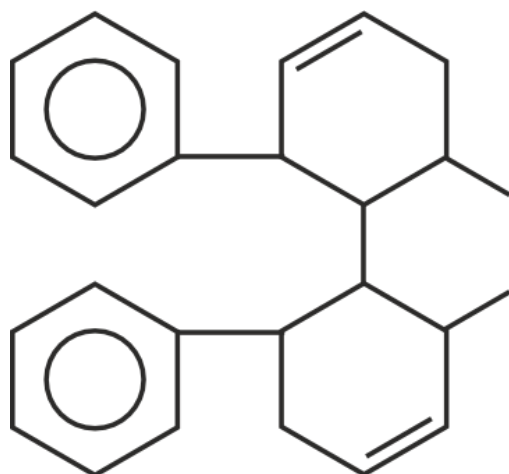
B



C

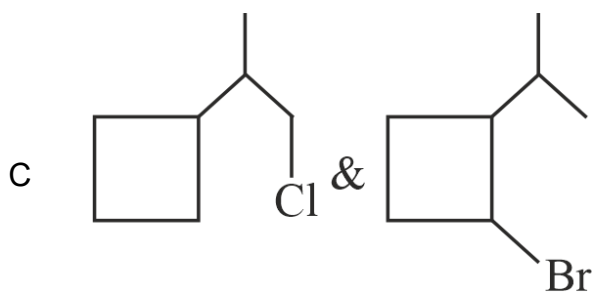
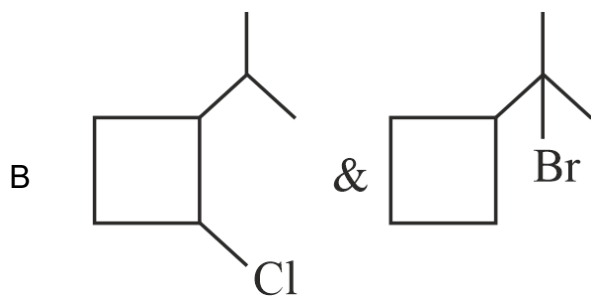
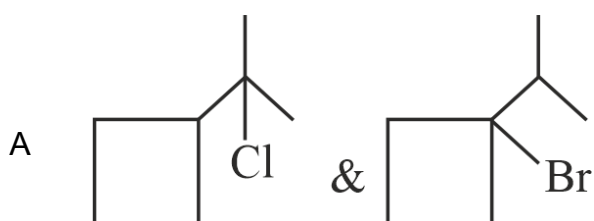
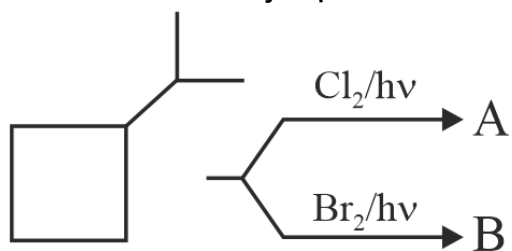


D

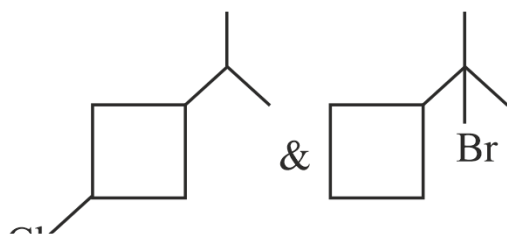


Question 66

When the given compound is subjected to chlorination and Bromination, what are the major product, A and B respectively?

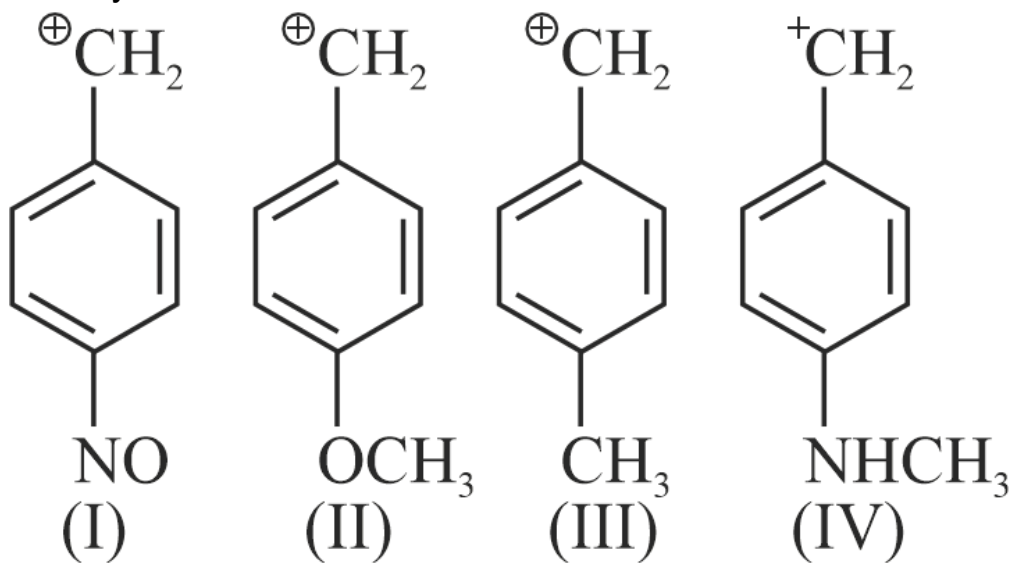


D



Question 67

Arrange the following carbocation in order of decreasing order of their stability.



A $\text{I} > \text{III} > \text{II} > \text{IV}$

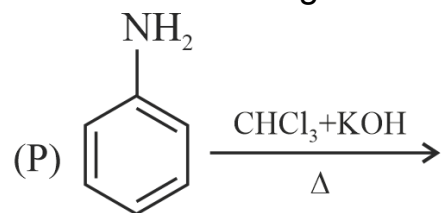
B $\text{IV} > \text{II} > \text{III} > \text{I}$

C $\text{III} > \text{II} > \text{IV} > \text{I}$

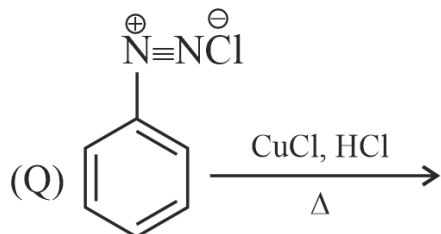
D $\text{IV} > \text{I} > \text{III} > \text{II}$

Question 68

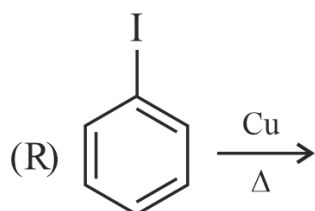
Match the following



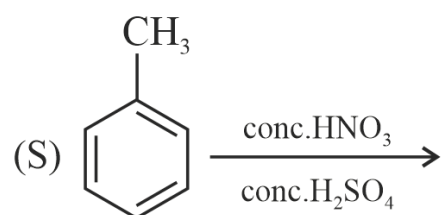
(1) Aryl halide is obtained as a product.



(2) It is electrophilic substitution reaction



(3) Reaction is isocyanide test



(4) New carbon-carbon bond is formed in product

A P Q R S
 4 1 2 3

B P Q R S
 3 1 4 2

C P Q R S
 1 2 4 3

Question 69

How many of the following statement is incorrect when a mixture of NaCl and $\text{K}_2\text{Cr}_2\text{O}_7$ is gently warmed with conc. H_2SO_4 ?

A A deep red vapour is evolved

B The vapours when passed into NaOH solution gives a yellow solution of Na_2CrO_4

C Chlorine gas is evolved

D Chromyl chloride is formed

Question 70

Nitrogen and Phosphorus are both non-metallic elements in Group 15. While Phosphorus can form compounds like PCl_5 and PCl_6^- , Nitrogen's highest observed covalency is different. Additionally, the $\text{N} - \text{N}$ single bond is significantly weaker than the $\text{P} - \text{P}$ single bond. What is the maximum covalency of Nitrogen, and what is the primary reason for the weaker $\text{N} - \text{N}$ bond compared to $\text{P} - \text{P}$?

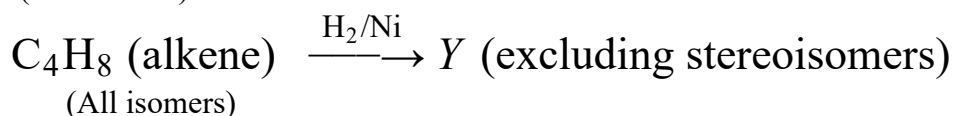
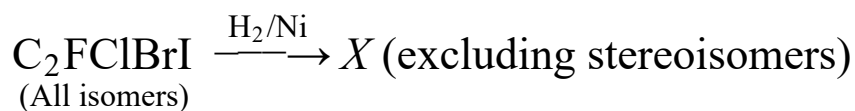
A 3; Larger atomic size of Nitrogen

B 4; High electronegativity of Nitrogen

C 4; Inter-electronic repulsion between lone pairs on adjacent Nitrogen atoms

D 5; Absence of d-orbitals in Nitrogen

Question 71



Find the value of $X + Y$.

Question 72

38 gm of mixture of Li_2CO_3 & MgCO_3 on strong heating showed a weight loss of 22 gm. Find the mole % of MgCO_3 in the given mixture (nearest integer).

Question 73

The standard enthalpy of combustion of $\text{C}_2\text{H}_6(\text{g})$ is $-1564.25 \text{ kJ mol}^{-1}$. Standard heat of formation of $\text{CO}_2(\text{g})$ and $\text{H}_2\text{O}(\text{l})$ are -395 kJ mol^{-1} and -286 kJ mol^{-1} respectively. The absolute value of heat liberated in kJmol^{-1} for the formation of 1 mole of $\text{C}_2\text{H}_6(\text{g})$ from its constituent elements in standard state is (nearest integer).....

Question 74

Identify the number of molecules/ions from the list below that have a linear molecular structure:

SO_2 , NO_2 , O_3 , ICl_2^- , SnCl_2 , H_2S , CO_2 , BeH_2 .

Question 75

In the molecular structure of $\text{Fe}_2(\text{CO})_9$, The difference between the number of triply bonded $\text{C} \equiv \text{O}$ units and doubly bonded $\text{C} = \text{O}$ units are/is how many.